Exercise sheet: Week 5

1. Let $\underline{X}$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$ and suppose it is required to test

$$H_0 : \mu = 0, \sigma = 1 \text{ v. } H_1 : \mu = 1, \sigma = 2.$$

Show that the critical region of the N-P MP test is

$$R = \{\underline{x} : 3\sum x_i^2 + 2\sum x_i \geq k\}.$$

2. Investigate the existence of UMP tests of one-sided hypotheses for the indicated parameters in the following distributions:
   a) μ in $\mathrm{Poi}(\mu)$,
   b) p in $\mathrm{Bin}(n, p)$,
   c) p in $\mathrm{Geo}(p)$.

3. [Randomised hypothesis test]

   Let $\underline{X}$ be a random sample from $\mathrm{Poi}(\mu)$ and suppose it is required to test

$$H_0 : \mu = 2 \text{ v. } H_1 : \mu > 2$$

   at a level of significance α.

   a) Use your conclusion from Question 2(a) to show that the critical region of the UMP test is

$$R = \left\{\underline{x} : \sum x_i \geq k\right\}$$

   where k is adjusted to achieve size α.

   b) Now assume that the sample size $n = 10$. By first deducing the distribution of $\sum X_i$, find the probability of a Type-I error, i.e., $\mathbf{P}(\underline{X} \in R | H_0)$, when
      i. $k = 27$
      ii. $k = 28$
      iii. $k = 28.6$
      iv. $k = 29$

      You may use R to perform the calculations.

      Is it possible to determine a value of k such that the Type-I error probability is exactly 5%?

   Let us now introduce the concept of a randomised hypothesis test. We denote by $d(\underline{X})$ a decision function whose values are terminal decisions (reject or fail to reject H_0). A randomised hypothesis test is governed by the following rule:

$$d(\underline{X}) = \begin{cases} 1 & \text{if } \sum X_i > k \\ 0 & \text{if } \sum X_i < k \\ p & \text{if } \sum X_i = k \end{cases}$$

Here, $d(\underline{X}) = 1$ means to reject H_0, $d(\underline{X}) = 0$ means to not reject H_0, whereas $d(\underline{X}) = p$ means to reject H_0 with probability $p \in (0, 1)$. One way to "implement" $d(\underline{X}) = p$ is to toss a biased coin, and reject H_0 if the coin lands on a 'Head', do not reject H_0 otherwise. We can choose the biased coin such that it lands on 'Head' with probability p, and lands on 'Tail' with probability $1 - p$.

The critical region is defined as follows:

$$R = \left\{\underline{x} : \left\{\sum x_i > k\right\} \cup \left(\left\{\sum x_i = k\right\} \cap A\right)\right\}$$

where A is the event of obtaining a 'Head' from tossing the biased coin.

c) Using the partition theorem, or otherwise, show that the probability of a Type-I error is equal to

$$\mathbf{P}\left(\sum X_i > k \mid \mu = 2\right) + p \times \mathbf{P}\left(\sum X_i = k \mid \mu = 2\right)$$

d) Using the results from parts (c) and (d), determine the values of k and p such that the randomised test attains $\alpha = 0.05$.

4. [Derivation of the common t-test]

Let $\underline{X}$ be a random sample from $\mathrm{N}(\mu, \sigma^2)$, where σ is unknown. For the GLRT of $H_0 : \mu = \mu_0$ v. $H_1 : \mu \neq \mu_0$, show that

$$\lambda(\underline{x}) = \left[1 + \frac{1}{n-1}\{t(\underline{x})\}^2\right]^{-\frac{n}{2}}$$

where $t(\underline{x}) = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}$, and hence that the critical region is

$$R = \{\underline{x} : |t(\underline{x})| \geq k\}.$$

(*Hint*: you will need to use $L(\hat{\mu}, \hat{\sigma}^2)$)
What is the k for the test of size α?